

WILDFIRE DETECTION AND MONITORING SYSTEM

Assessment 2 Part A

Systems Definition & Functional Hierarchy Report

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Systems Engineering Principles

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EXECUTIVE SUMMARY

This report outlines a wildfire detection and monitoring system designed to reduce time difference between an incipient flame igniting and first responders being alerted. This system fuses data gathered from satellites and UAV patrols which is the autonomously screened with flagged anomalies being manually confirmed at the operation hub. Once the fire is confirmed an alert is transmitted to fire departments and the public containing the fire's geolocation and estimated coverage, whilst the system continues to monitor the fire and provide live updates to stakeholders. This report defines the system through applying the V-lifecycle, starting at stakeholder needs then CONOPS, followed by user requirements, system requirements and then the functional and physical architecture. The report contains defined stakeholders, CONOPs analysis, quantifiable target requirements, architecture for 7 subsystems and 8 functions, FFBDs and system KPIs such as detecting fires before they cover 5 acres and fore stakeholders to receive a confirmed alert within 10 minutes of ignition. The report following a systems engineering approach produced a design which is integrated and verifiable with the system meeting all outlined objectives.

INTRODUCTION

BACKGROUND

CSIRO calls Australia "One of the most fire-prone countries on Earth." The damages resulting from Australia's bushfires are not only limited to millions of dollars' worth of infrastructure but also have a tremendous impact on the vast population of humans, wildlife and vegetation. (CSIRO n.d.)

Black Summer was Australia's worst fire season in recent years, burning 17 million hectares of land. It devastated every state in the country leaving behind 3094 destroyed homes, killed 33 people and it is estimated over 1 billion animals perished. The IRP state that the "recovery and restoration of the native flora and fauna need and annual investment of \$16 billion to avoid extinction risks. (UN Office for Disaster Risk Reduction [UNDRR] 2025)

With an increase in the frequency of bushfires, the aftermath of Black Summer is an indication for the necessity of improved detection and monitoring systems. With proper implementation, they provide more opportunities for stakeholders to battle the fires destruction.

PROBLEM STATEMENT

The current detection and monitoring systems in place identify blazes before they take off are riddled with issues such as being too slow or not covering enough area, leading to delayed response times. To be successful the system outlined in this report will need to confirm there is a fire within minutes of it starting, day or night, and convey that info to the necessary stakeholders.

PROJECT OBJECTIVES

This system must be able to detect fires before it covers an area of 5 acres. According to reports for the Victorian royal commission, “more than 80% of fires are contained as before they can grow” when they are less than 5 hectares (“2009 VBRC - Final Report - Vol 2 - Fireground Response” n.d.). This goal of 5 acres (≈ 2.02 hectares) allows time for firefighters to arrive at the location before the fire grows too much. This system should also have a false alarm rate under 5% as well as supply a geolocation tag within 300m from the detection point. It must also be freely available for the population so they have the most up to date information available.

PROJECT SCOPE

This system will include detection, confirmation, location, impacted area, tracking and predictions. This system won't be designed to help physically combat the fire or provide logistics to the fire department to assist in combatting the fire.

ASSUMPTIONS & CONSTRAINTS

In this report the assumption is made that this is a unilateral effort with nations providing satellite feeds available from their space agencies such as NASA and the ESA. Other detection methods may also be utilised such as UAV's, which have a limited battery capacity, limiting the area and duration they can be active. There are also budget constraints as this being a free to access service, it is likely government funding is needed. In this report it is also assumed that the system will be most active during and around the summer season, so in theory more power can be generated from solar. This allows for battery powered methods of detection to be more effective in covering areas which are don't have access to the electrical grid.

SYSTEM CONTEXT ANALYSIS

SYSTEM CONTEXT

Fires are most prone to start at locations with high fuel loads and dry vegetation such as grasslands and dense woodlands. These locations are commonly found on the fringes of cities and towns where housing development ends. These locations also pose a high level threat being so close to dwellings, so the system will be most prevalent in these locations. It will also have to operate in all weather conditions at all times of the day. It will use external systems to receive information such as NASA satellite feeds and BoM weather data. It will also need to connect with emergency responders and the population to convey vital information. Using UAVs in the system will also have to comply with regionalised airspace rules, particularly in locations close to residential dwellings.

STAKEHOLDER IDENTIFICATION, ANALYSIS & NEEDS

Table 1, Stakeholder Identification, Viewpoints & Primary Needs

STAKEHOLDER	VIEWPOINT	PRIMARY NEED
Fire departments	Primary Users	Cover a wide area and receive an accurate alert as early as possible.
Firefighters	Responders	Accurate location, behaviour prediction, works any time.
General Population	At risk	Receive evacuation warnings before it becomes too dangerous to leave, up to date information on fires.
Government	Funding, regulator	A reliable service that can cover locations all over the country.
Aviation regulators	Regulator	UAV operations don't put any of the population at risk and are compliant with the current regulations.
Systems maintenance and operation crew	System support	Reliable, maintainable and up to date supplies.
Landowners and insurance brokers	Asset management	Protect their properties from damages caused by blazes.

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STAKEHOLDER PRIORITISATION

Table 2, Stakeholder Prioritisation Table

	Fire departments (CFA)	Firefighters	General Population	Government	Aviation regulators	Systems maintenance and operation crew	Landowners and insurance brokers	SCORE
Fire departments (CFA)		1	1	1	1	1	1	6
Firefighters	0		1	1	1	1	1	5
General Population	0	0		1	1	1	1	4
Government	0	0	0		1	1	1	3
Aviation regulators	0	0	0	0		0	1	1
Systems maintenance and operation crew	0	0	0	0	1		1	2
Landowners and insurance brokers	0	0	0	0	0	0		0

Table 3, Stakeholder Prioritisation Ranking

RANK	STAKEHOLDER
1	Fire departments (CFA)
2	Firefighters
3	General Population
4	Government
5	Systems maintenance and operation crew
6	Aviation regulators
7	Landowners and insurance brokers

CONTEXT OF USE

Normal operations occur when there is no fire present. The detection systems will provide continuous surveillance.

Adverse conditions may occur during the night, under heavy smoke or in poor weather. In these scenarios the system will continue to operate to a point it is still safe for all stakeholders or to where it is unable to complete its purpose. If it becomes unsafe then alternate detection methods may come into play. E.g. smoke blocking satellite vision, UAVs deployed as a replacement.

Emergency conditions such as when multiple fires start or if there is infrastructure outages may strain the system, however it can be managed to minimise non-optimal operations. If there are multiple fires in a location, the detection system will perform the objectives before moving onto the next fire, providing as much information to stakeholders as possible. Additionally, with infrastructure outages different methods of surveillance can be used, such as switching from satellite detection to UAVs.

In the future it can be assumed that there will be an increased number of satellites with higher resolution imaging of earth which if utilised will lead a timelier and more accurate detection system. UAVs will also most likely have improved battery capacity and reduced costs, allowing for a greater number to swarm areas and providing faster detection. AI can also be implemented to identify fires instead of it being done manually.

OPERATIONAL SCENARIOS

ROUTINE MONITORING

During routine monitoring during a day classified as high fire danger risk satellites pass over locations which are monitored for any anomalies, baseline images are captured for location comparison purposes, UAV's patrol highest risk areas such as grasslands near dwellings to provide the earliest possible detection. If no fires start surveillance continues.

INCIPIENT FIRE DETECTION & CONFIRMATION

During surveillance a thermal anomaly is flagged by a satellite passing over the area or by a UAV, fire departments are alerted to a potential fire and to prepare. Running parallel to this, a person starts checking if it is a false flag through imaging checked against recent historical imaging of the area and visual confirmation before a false positive is given. If the fire is confirmed, then it will be geolocated and mapped with this information being immediately communicated back to the fire department who have already prepared to head out. Additional UAVs are dispatched to confirm and gather more info for firefighters, fire departments and the population.

POOR CONDITION DETECTION

Under conditions such as thick smoke during the night where normal camera imaging isn't providing adequate data, the system relies on thermal imaging to detect thermal anomalies where the temperature captured ranges from 600°C to 1100°C, the heat at which bushfires typically reach (CSIRO n.d.). A person then confirms the thermal imaging is consistent with a fire, where the standard operating procedures take place and the info is then convey to the stakeholders.

EXPECTED SYSTEM OUTCOMES

ROUTINE MONITORING

During routine operations it is expected that in majority of surveillance ops nothing occurs. This would mean the area continues to be monitored until resources are reallocated to a higher risk area. With UAV surveillance when the drone's battery runs low, it will be grounded and swapped for a new battery or charged whilst a different UAV is released into the sky.

INCIPIENT FIRE DETECTION & CONFIRMATION

After an incipient fire is detected and the outlined stakeholders are alerted before the fire grows to 5 acres. The additional UAVs detected to monitor the fire and the surrounding areas will continue to give mapping data and forecasts as well as look out for any additional incipient fires that have started because of the first. If there are these fires they will also be geolocated and mapped with the information given to fire departments and the public.

POOR CONDITION DETECTION

Once the identification of a fire is communicated to the relevant parties, more UAV's will be launched to patrol the area for continued mapping of the fire as well as for detection of other fires starting up nearby. These details will then continue to be communicated to relevant parties.

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CONCEPT OF OPERATIONS (CONOPS)

OPERATIONAL PHILOSOPHY

The primary user of this service are fire departments, who will need accurate information in a timely manner. Upon a potential detection of a fire, they will receive notification that they should start preparing their gear whilst the confirmation check is performed. If the system then confirms the threat, the department will receive the location and mapping of the fire. Once at the location the departments will continue to receive live updates on the condition of the fire as well as surrounding factors to consider such as housing or other incipient fires which they can then convey to the appropriate personnel.

MAIN CONCEPT OF OPERATION

1. Sensors operating autonomously acquiring data through satellite imaging, thermal imaging and on days where there is a high fire risk, UAVs.
2. Data is then transmitted to either localised or national operations centre.
3. This data is then screened with a trained AI to detect any anomalies, which are then flagged for confirmation.
4. The flagged anomaly communicated to the relevant fire departments to keep them informed on a potential blaze.
5. Flagged anomalies are then checked by a person at the operation hub to determine whether it was a false flag or a real threat.
6. Once the anomaly has been confirmed as a fire, the location and initial estimate coverage is then given to the fire department to act on.
7. Whilst the fire department is being informed of the fire, a swarm of UAVs will be launched to monitor the fire's development to provide up to date information on its size, location and other small fires starting up nearby to the relevant parties.
8. The UAVs will continue to monitor the fire until it becomes controlled, it burns out or until they run out of power, where a new fleet of UAVs would be dispatched to allow the old ones to recharge.

ALTERNATIVE CONCEPTS OF OPERATION

Other modes the system may have to operate in may be the loss of communication during a fire, where the UAVs are unable to transmit their data back to the base of operations which may be caused by destroyed cell towers. In this scenario the system can run a basic mode using satellite data to survey the area. However, due to the current assumption of using NASA's satellites which sit in low earth orbit, they can only survey a location for 10 to 15 minutes before it passes. Additionally, whilst this is occurring, operators of the UAVs will set up alternate methods of communication between the on scene personnel and the operations hub through SATCOM channels. Furthermore, if the fire is impacting the connection between the UAV's and operators, fibre optic cables can be applied to the drones for perimeter surveillance.

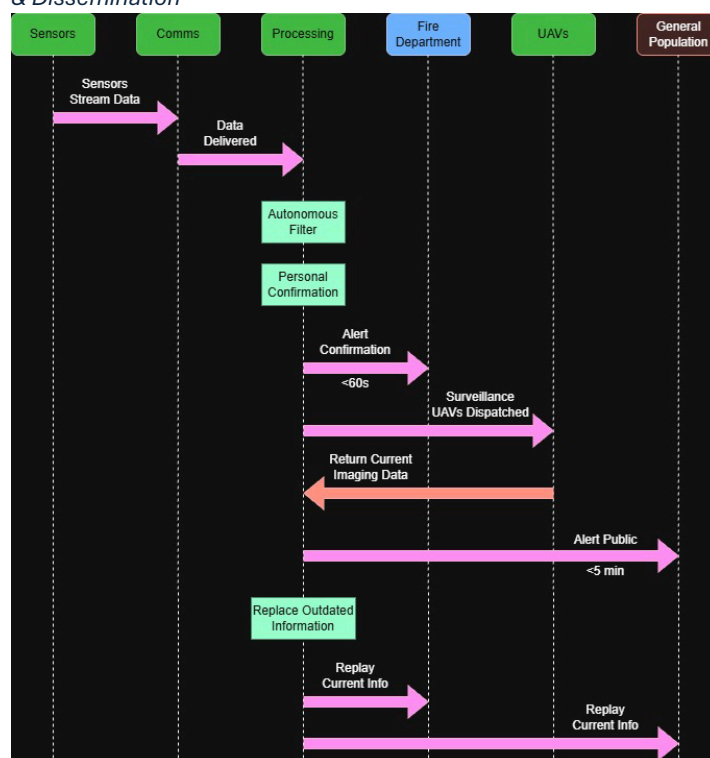
Another backup mode which the system can operate in is the loss or failure of the AI powered automatic screening. In the case where this fails, the job can be assumed by a person or team

to complete manual checks of the data collected in real time. This, whilst being resource intensive, will allow for the system to continue to operate adequately.

An alternative mode for this system could be for UAV operations in remote dead zones, where the data obtain from surveillance can't be transmitted back to the main hub for analysis. In this scenario the team would have access to a local initial screening AI. Then when a fire is confirmed, using SATCOM, the information can be conveyed to the relevant parties as well as the operations hub, where they can continue analysis with resources such as satellites or send another team with more UAVs to increase surveillance of the area.

OPERATIONAL SEQUENCE DIAGRAMS

Figure 1, Operational Sequence Diagram for Detection, Confirmation & Dissemination



USER PERSPECTIVE ANALYSIS

The operations hub will have a screen outlining which areas are being monitored at any given time. In the case of a flagged anomaly, operators will be given the data to confirm the autonomous screening's conclusion. Once the operator confirms that the anomaly is a fire, that information is then sent out to the relevant parties. The operators will then notify the appropriate people in the building to initiate the system's next steps. This includes launching a swarm of UAVs to the location to continue monitoring the situation and relaying further information. Once one of the requirements outlined in *step 8* under the *main concept of operation* the UAVs will be returned to their initial location and surveillance will continue as normal.

Fire departments using this system will receive notification of a potential fire at a given location, giving them time to prepare. Once the operations hub has confirmed the fire, the department will

receive further communications ratifying the earlier fire's validity, the location and the area it covers. From here the firefighters will begin fighting the blaze, where the department will continue to receive live analytics from UAVs surveilling above.

Residents using this system will be notified of any relevant fires once they it has been confirmed by operators. They will be provided with the same information as the fire departments so the community is fully informed. From this they can make a decision on their next course of action.

INNOVATIVE OPERATIONAL FEATURES

An improve feature to the system may be to for the UAV to host the autonomous screening, improving efficiency through reducing the delay in communication. Along with this may also be the implementation of AI driven UAVs, reducing the labour resources required for operation. Through the implementation of autonomous UAVs, the system can vastly increase the detection coverage as they don't need as many manual operators.

For an indefinite and large area coverage, the system can implement the use of geostationary satellites. These satellites remain stationary to a geolocation of earth's surface, eliminating blind spots created after passing satellites have left an area. This may be through using another party's satellite feed or launch satellites solely for the systems operations for precise targeting of specified locations.

USER REQUIREMENTS

USER NEEDS

We need to make a system that accurately detects wildfires and can communicate that information to firefighting services and the public in a timely manner. It also needs continue to provide information regarding the wildfire and its actions to these same parties.

REQUIREMENT DEVELOPMENT PROCESS

Each user requirement outlined in the *user requirement table* below has been derived from an analysis of the CONOPS section of the report, following the stakeholder needs step of the SE V-lifecycle. The user requirements were found by looking at the *stakeholder prioritisation* table, the *context of use*, the *operational scenarios* and both the *main and alternative CONOPS concepts*. The needs of high level stakeholders moulded the CONOPS, which is used to produce the overarching user requirements. As a result of completing the CONOPS, needs were discovered that weren't outlined in the initial stakeholder needs such as with the implementation of thermal imaging. To derive the user requirements from CONOPS, for each step in the systems operations, what a user must do or have needs to be inferred by the engineer. The user requirements outlined need not to contain solutions and have system specifications excluded into system requirements. Potential user requirements are then consolidated to eliminate any duplicated, noting any conflicts and assigning each an unique identifier.

USER REQUIREMENT TABLE

Table 4, User Requirement Table inc. Sources & Justifications

UR ID	USER REQUIREMENT	SOURCE	JUSTIFICATION
UR-01	Be notified of any fires as early as possible.	Fire Departments	Smaller fires have much higher chances to be controlled.
UR-02	Receive accurate information of the fire's location and coverage.	Fire Departments/ Step 6	Know its exact location and size help departments to plan.
UR-03	Coverage over large areas.	Fire Departments/ Government	By covering a large area, it is more likely a fire can be identified before it grows too large.
UR-04	Up to date details about the fire.	Fire Departments & Public	Drives responses and resource allocation.
UR-05	Receive timely warnings.	Public	Better chance to make a safe evacuation.
UR-06	UAV operations to comply with aviation regulations.	Aviation Regulators	Safety and legal mandate.
UR-07	The system to be reliable and easily maintainable.	Operators/ Maintenance Crew	If the equipment can't be maintained the system can't complete its task to an adequate standard.
UR-08	Protect assets and infrastructure from damages.	Property Owners/ Insurance Brokers	Large economic losses impact both individual and state fund allocations.
UR-09	Confidence that flagged anomaly is a fire and not a false alarm.	Fire Departments/ Public/ Step 3/ Step 5	False flags waste resources and degrades trust in the system.
UR-10	Detection for secondary fires which may sprout from the initial fire.	Step 7	Poses a safety threat to people in the area if it goes unnoticed.
UR-11	The system to continue to operate under the loss of some vital sections.	Backup Concept	If the system fails then stakeholders can't access important information.
UR-12	The system to be able to track multiple fires in the same location	Context of Use	A fire starting other fires is common and not being able to track those poses a safety threat to those nearby.
UR-13	The system to detect fires through smoke and during the night	Poor Condition Scenario	Fires can start anytime and optics aren't as effective during the night or if the line of sight is being impeded.

REQUIREMENT PRIORITISATION

Table 5, User Requirement Prioritisation Table

	UR-01	UR-02	UR-03	UR-04	UR-05	UR-06	UR-07	UR-08	UR-09	UR-10	UR-11	UR-12	UR-13	Score
UR-01		1	1	1	1	1	1	1	1	1	1	1	1	12
UR-02	0		1	1	1	1	1	1	1	1	1	1	1	11
UR-03	0	0		1	1	1	1	1	0	1	1	1	0	8
UR-04	0	0	0		0	0	0	1	0	1	1	1	0	4
UR-05	0	0	0	1		1	1	1	0	1	1	1	0	7
UR-06	0	0	0	1	0		0	1	0	1	1	1	0	5
UR-07	0	0	0	1	0	1		1	0	1	1	1	0	6
UR-08	0	0	0	0	0	0	0		0	1	0	1	0	2
UR-09	0	0	1	1	1	1	1	1		1	1	1	1	10
UR-10	0	0	0	0	0	0	0	0	0		0	0	0	0
UR-11	0	0	0	0	0	0	0	1	0	1		1	0	3
UR-12	0	0	0	0	0	0	0	0	0	1	0		0	1
UR-13	0	0	1	1	1	1	1	1	0	1	1	1		9

Table 6, User Requirement Ranking

RANK	UR IDENTIFIER
1	UR-01
2	UR-02
3	UR-09
4	UR-13
5	UR-03
6	UR-05
7	UR-07
8	UR-06
9	UR-04
10	UR-11
11	UR-08
12	UR-12
13	UR-10

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USER REQUIREMENT CONFLICT ANALYSIS

UR-01 and UR-09 have conflicting requirements as alerting the fire department as soon as possible requires more sensitive autonomous detection which lead to a greater number of false positives. This contradicts the user requirements to have confidence in its classification of anomalies detected which needs to be more thorough. To fix this, increased resources could be reallocated to the second manual screening of detected anomalies and increase the sensitivity for faster detection times whilst minimising false alarms where fire department units are dispatched.

Another is UR-03 and UR-14, where to cover countries such as Australia requires immense resources, which runs up the costs to launch and operate. A potential way to solve this issue is a large onetime spend on enough geostationary satellites to cover the area required. This whilst having lower running costs than purchasing a fleet of UAVs to cover the same area and their maintenance.

REQUIREMENT VALIDATION AND FEASIBILITY DISCUSSION

All the user requirements outlined are feasible with current technologies. UAV development R&D has skyrocketed since the start of the Russo-Ukrainian War which is likely to lead to UAVs more capable of covering larger areas for longer periods of time at lower costs. This will allow the system to cover more of a nation, increasing the number of fires which can be detected before they become too big to contain.

Additionally, satellite imaging is rapidly improving, showing better resolution captures of earth's surface inviting the opportunity for vast area detection over the most remote parts of counties where access with UAVs and personal is far more difficult. Soon there will be the potential to detect all incipient fires that start preventing most bushfires from occurring, however there will be urban and suburban areas where it wouldn't be too ethical to gather data.

SYSTEM REQUIREMENTS

REQUIREMENT DEVELOPMENT METHODOLOGY

Each of the user requirements outlined in the above section were converted to a system requirements in the form of 'shall' statements, defining what the system must do. To convert user requirements into system requirements this report used SMART wording and the INCOSE requirement rules, allowing for requirements to be measurable and verifiable. For each of the outlined user requirements, questions posed such as "what must the system achieve?" and "what is the quantifiable threshold needed to achieve this?", converting qualitative needs to quantitative ones. User requirements will produce one or more system requirements, each given a unique identifier so it can be linked back. The *Requirement Traceability Matrix* subsection of this report below records the link from user requirements to system requirements.

FUNCTIONAL REQUIREMENTS

Table 7, System Requirements inc. Sources & Verification Methods

SR ID	SYSTEM REQUIREMENT	SOURCE	VERIFICATION
SR-01	This system shall communicate a confirmed fire with a geolocation tag within 10 minutes from ignition in high risk zones.	UR-01/ UR-02/ UR-05	Test response time on a timed controlled ignition.
SR-02	The system shall be capable of detecting fires at a minimum size of 25 m ² .	UR-01	Test optics on controlled burns.
SR-03	This system's autonomous detection shall achieve a positive verification rate of 90% or more.	UR-09	Test against existing imaging.
SR-04	The system shall achieve 95% positive verification on issued fire alerts to stakeholders.	UR-09	Test both autonomous detection and manual checks against existing imaging.
SR-05	The system shall be capable of detecting fires after dark and through smoke.	UR-13	Test equipment at night and with the vision obscured.
SR-06	The system shall be able to map the coverage of the fire a keep it updated with an accuracy of 90%	UR-02, UR-04	Test on controlled burns with a controlled and known area.
SR-07	The system shall be capable of covering all dedicated high risk areas at all times the system is operational.	UR-03/ UR-08	Compare coverage of an area at a given time to the total observation area.
SR-08	This system shall comply with UAV regulations set by CASA or any other aviation authority in each region.	UR-06	Inspections from aviation authorities.
SR-09	This system shall have regular crews able to perform maintenance on 80% of its equipment out in the field.	UR-07	Teach crew maintenance needs so they can perform it out in the field.
SR-10	This system shall have backup processes that can replace 90% failures that occur.	UR-11	Test backup process against planned failures.
SR-11	The system shall have a swarm of UAVs patrolling 99% of the area surrounding the existing fire to monitor for additional blazes.	UR-10/ UR-12	Test in a controlled burn where there are multiple ignition points.

NON-FUNCTIONAL REQUIREMENTS

Table 8, Categorise Non-Functional Requirements

PERFORMANCE	RELIABILITY	SAFETY	SUSTAINABILITY	MAINTAINABILITY
SR-01	SR-03	SR-07	SR-09	SR-09
SR-02	SR-04	SR-08		
SR-05	SR-07	SR-10		
SR-06	SR-09	SR-11		
SR-07	SR-10			
SR-10				
SR-11				

REQUIREMENT TRACEABILITY MATRIX

Table 9, Requirement Traceability Matrix from User Requirements to System Requirements

		TRACES TO										
		SR-01	SR-02	SR-03	SR-04	SR-05	SR-06	SR-07	SR-08	SR-09	SR-10	SR-11
TRACES FROM	UR-01											
	UR-02											
	UR-03											
	UR-04											
	UR-05											
	UR-06											
	UR-07											
	UR-08											
	UR-09											
	UR-10											
	UR-11											
	UR-12											
	UR-13											

LEGEND

TRACED

NOT TRACED

COMPLIANCE WITH ENGINEERING PRINCIPLES

Table 10, Compliance of System Requirements inc. SMART Principles

SMART CRITERIA	REQUIREMENT COMPLIANCE
Specific	Detect and confirm the presence of a wildfire whilst still small
Measurable	Fire coverage < 5 acres, false alarm rate < 5%
Achievable	It is feasible with UAV and geostationary satellite monitoring implementing thermal imaging and autonomous screening
Relevant	Satisfies UR-01 and fulfills the primary objective of the report
Time-bound	Fires have a much greater chance of containment before exceeding 5 acres of coverage

All system requirements outlined in the *Functional Requirements* section of this report were written to fulfill SMART and INCOSE method rules.

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VERIFICATION MEASURES

Table 11, System Requirement Verification Measures

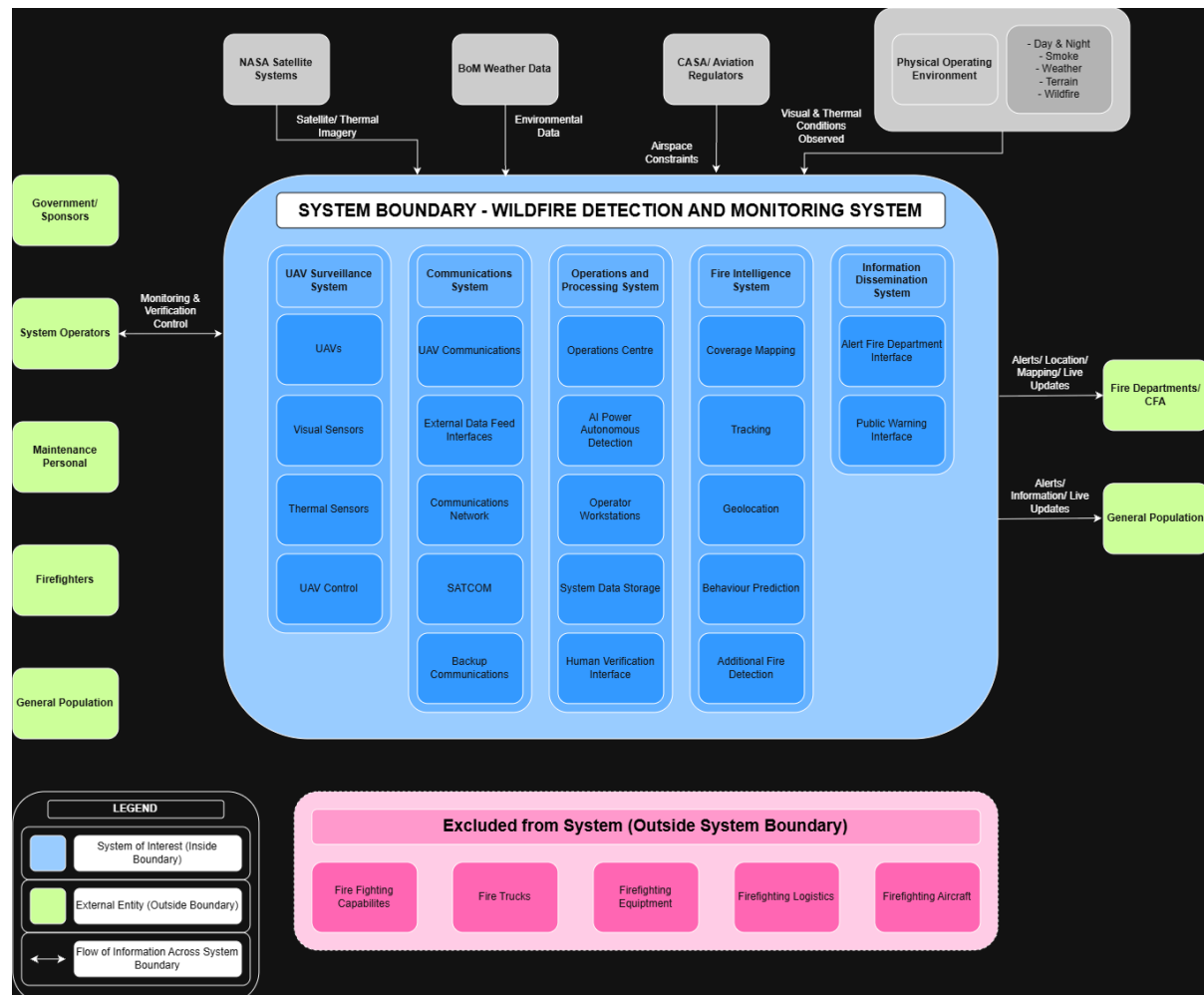
SR ID	VERIFICATION
SR-01	Test response time on a timed controlled ignition.
SR-02	Test optics on controlled burns.
SR-03	Test against existing imaging.
SR-04	Test both autonomous detection and manual checks against existing imaging.
SR-05	Test equipment at night and with the vision obscured.
SR-06	Test on controlled burns with a controlled and known area.
SR-07	Compare coverage of an area at a given time to the total observation area.
SR-08	Inspections from aviation authorities.
SR-09	Teach crew maintenance needs so they can perform it out in the field.
SR-10	Test backup process against planned failures.
SR-11	Test in a controlled burn where there are multiple ignition points.

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HIGH-LEVEL SYSTEM DESIGN

SYSTEM BOUNDARY DIAGRAM

Figure 2, System Boundary Diagram inc. Internal & External Elements of the System



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CONTEXT DIAGRAM

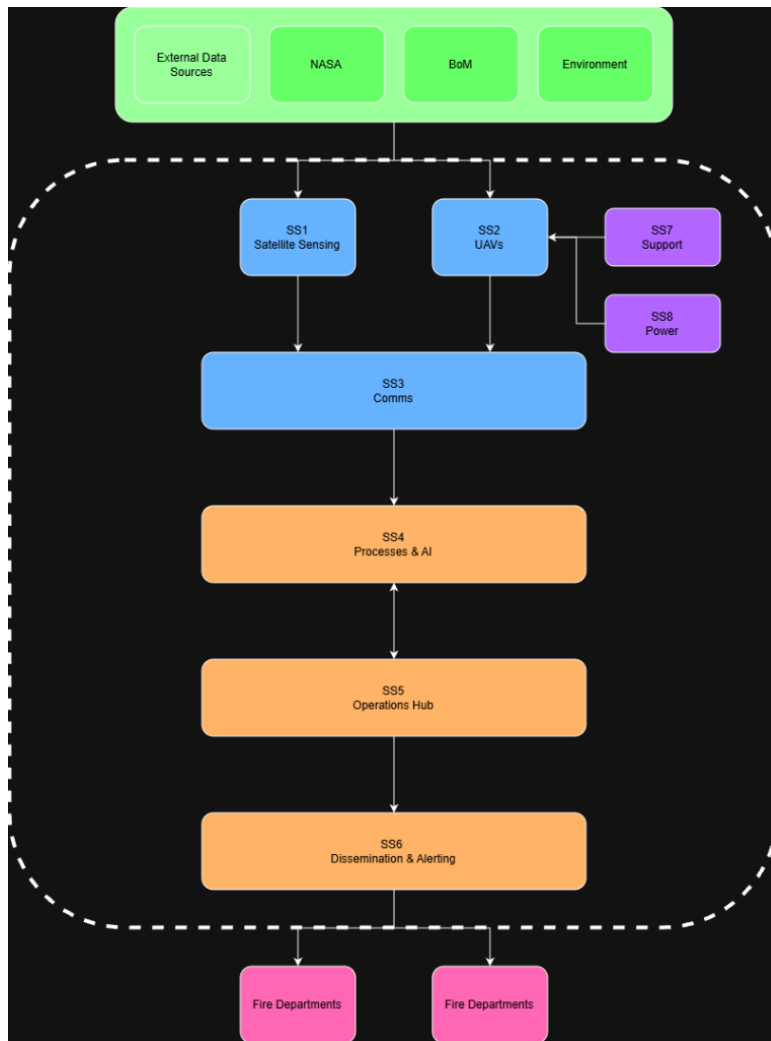
Figure 3, Context Diagram Outlining Data Flows & External Entities



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SYSTEM ARCHITECTURE

Figure 4, System High Level System Architecture



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MAJOR SUBSYSTEMS

Figure 5, Major Subsystems inc. Principal Components



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INTERFACE DESCRIPTION

Figure 6, N² Interface Matrix Displaying Data Transfer Between Subsystems



FUNCTIONAL ANALYSIS

FUNCTIONAL ARCHITECTURE OVERVIEW

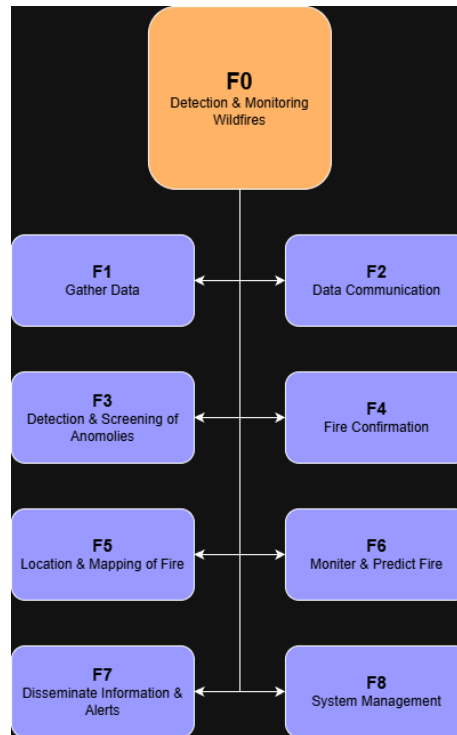
The systems top level function is to detect and monitor wildfires by taking sensor and environmental inputs and converting them into localised fire logistics and alerts. The eight functions operating in this system is Gather Data, Data Communication, Detection & Screening of Anomalies, Fire Confirmation, Location & Mapping of Fire, Monitor & Prediction Fire, Disseminate Information & Alerts, and System Management. Each function feeds into the next, for example, acquired data is given to communication networks which is then fed to detection methods. Without the different functions generating their output, the system wouldn't have the operation capabilities to complete its designated tasks. Each level 1 function can be decomposed into level 2 and sometimes level 3 sub functions which is outlined in the Functional Hierarchy Diagram in the following subsection. The functions are allocated to the subsystems listed in the High-Level System Design section of this report, connecting the functions to the given physical architecture.

Table 12, Function Allocation

Function Allocation	Name
F1	Gather Data
F2	Data Communication
F3	Detection & Screening of Anomalies
F4	Fire Confirmation
F5	Location & Mapping of Fire
F6	Monitor & Prediction Fire
F7	Disseminate Information & Alerts
F8	System Management

FUNCTIONAL HIERARCHY DIAGRAM

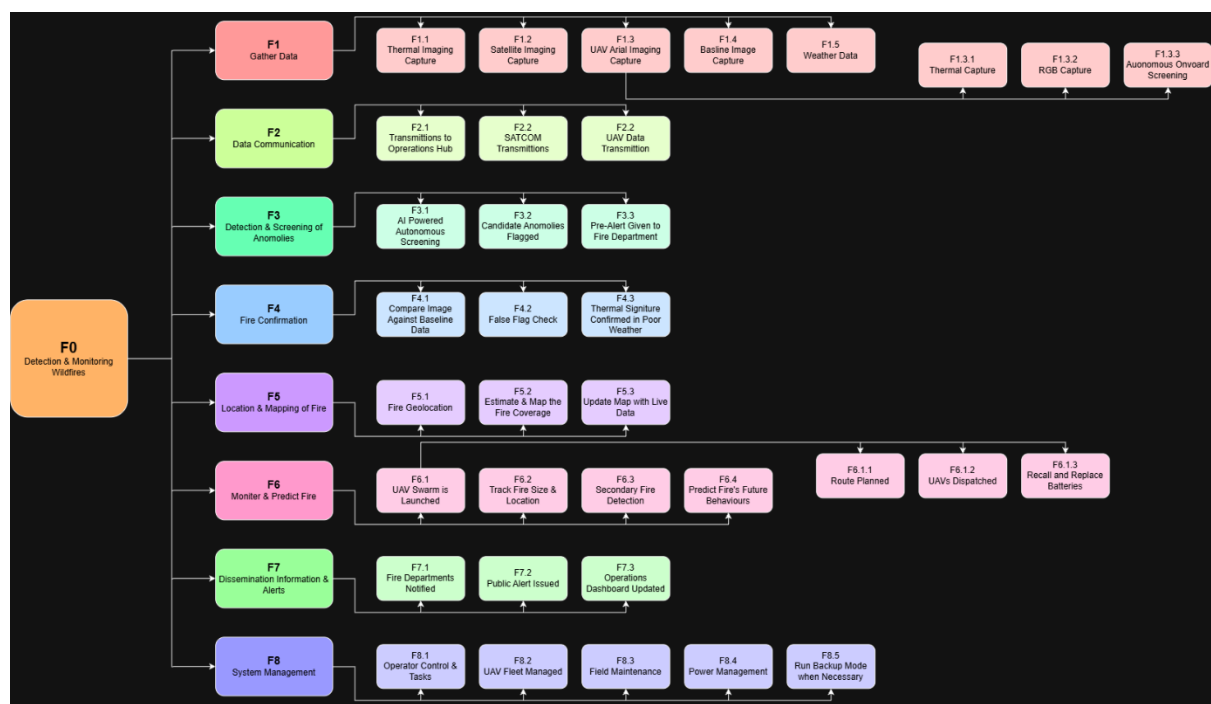
Figure 7, Level 0-1 Functional Hierarchy Diagram



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FUNCTIONAL DECOMPOSITION

Figure 8, Level 0-3 Functional Hierarchy Diagram



FUNCTION ALLOCATION TO SUBSYSTEMS

Table 13, Function Allocation to Subsystems

	SS1	SS2	SS3	SS4	SS5	SS6	SS7	
F1								
F2								
F3								
F4								
F5								
F6								
F7								Primary Responsibility
F8								Supporting Responsibility

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INPUT-OUTPUT ANALYSIS

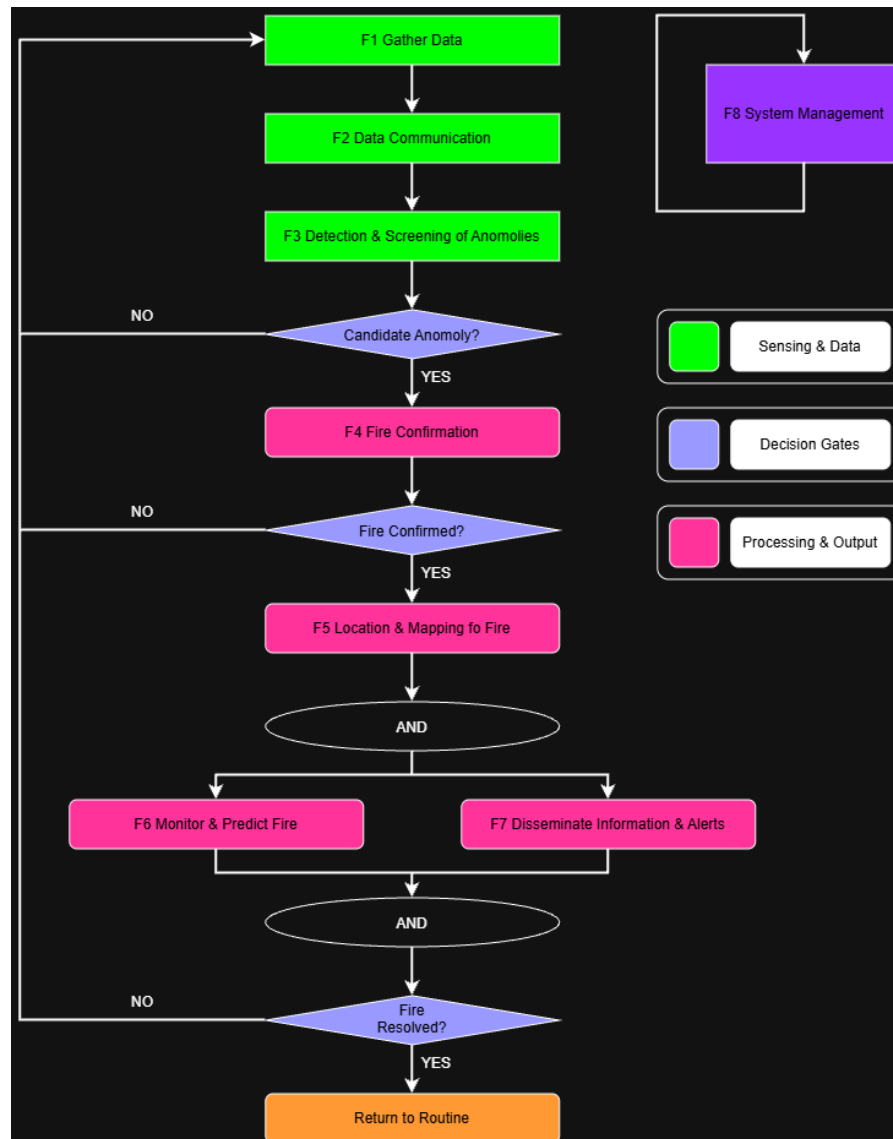
Figure 9, F1 → F8 Input-Output Analysis



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FUNCTIONAL FLOW BLOCK DIAGRAMS (FFBD)

Figure 10, Level 0 Functional Flow Block Diagram



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FUNCTIONAL INTERACTION ANALYSIS

Table 14, Functional Interaction Analysis inc. Data Contents Passed Between Functions

FUNCTION	SENDS TO	CONTENTS
F1	F2	Raw thermal and optical imaging, and sensor data.
F2	F3	Data transmitted to operations hub.
F3	F4	Anomalies flagged and pre-alert given.
F4	F5, if rejected $\cup$ F1	Fire confirmed or rejected.
F5	F6, F7	Fire geolocation and coverage mapping.
F6	$\leftrightarrow$ F4	Secondary fire, live mapping and spread forecast.
F7	External Systems	Fire department alerts, public alerts, info dashboard.
F8	F1:F7	Charged UAVs, backup mode operation, tasking.

SYSTEM INTEGRATION THROUGH FUNCTIONAL FLOWS

System integration can be demonstrated by following an operational thread through its linked functional flows and the data transmitted across each subsystem. The thread followed in this report starts at F1 and is then transferred as SS1 and SS2 to F2 where SS3 is then transmitted to F3. The data output represented by SS4 from F3 is then transferred to F4, which is then given by SS5 to F5. SS4 data is then transferred from F5 to F7 which then transmits SS6 to the fire department and the public. Each handoff each function gives to the next is defined and can viewed in the diagram under the *Interface Description* section of this report. This outlines how there are no function outputs which are left unused and will be utilised at some point in the systems flow. The unbroken chain from collecting raw data at the start of the systems operations to disseminating information to the relevant parties at the end is evident of the successful integration of subsystems into the entire system.

OPERATIONAL FLOW VERIFICATION

Table 15, Operational Flow Verification inc. Functional Flow Paths Mapped to Objectives

FUNCTIONAL FLOW PATH	OBJECTIVE OR REQUIREMENT VERIFIED
F1 $\rightarrow$ F2 $\rightarrow$ F3 $\rightarrow$ F4 $\rightarrow$ F5	Confirm before reaching an area of 5 acres and the exact geolocation of the fire.
F3 $\rightarrow$ F4	False alarm rate under 5%.
F5 $\rightarrow$ F7	Fire departments and the public receive timely alerts.
F6 $\leftrightarrow$ F1	Continuous monitoring of the fire until it is controlled or resolved.
F8 $\parallel$ Main Flow	System resilience and availability during flow.

PRELIMINARY SYSTEM INTEGRATION AND SUCCESS CRITERIA

EVIDENCE OF SYSTEM INTEGRATION

The evidence of system integration can be found in the diagrams under the System Architecture, Interface Description, Function Allocation to Subsystems, and Functional Flow Block Diagrams sections of the report. Every level 1 function has been allocated to a minimum of one subsystem with no function being left unused, additionally, every interface outlined in the diagram found in the Interface Description section has both a source and destination. Both operational and control paths are present in this system with SS4 and SS5 having manual work inserted into the flow for confirmations and SS5, the operations hub transmitting tasks back to UAVs at SS2, respectively. Because the operational and control paths are defined in the report, it outlines that the system doesn't act as a unidirectional data pipeline, but as a bidirectional. With each function having a respective subsystem, all interfaces being defined, and the unbroken flow of data, it suggests successful system integration.

KEY PERFORMANCE INDICATORS

Table 16, Key Performance Indicators, Targets & Sources

KPI ID	KPI	TARGET	SOURCE
KPI-01	Fire size when detected	< 5 acres	Project Objective & UR-01
KPI-02	False alarm rate	< 5%	Project Objective & UR-09
KPI-03	Detection response + Dissemination timeline	Confirm, geolocate and transmit alerts within 10 minutes of ignition	UR-01 & UR-02 & UR-05
KPI-04	Minimum detection size	25 m ²	UR-01
KPI-05	Autonomous detection accuracy	> 90% Accuracy	UR-09
KPI-06	Dissemination alert accuracy to stakeholders	> 95% Accuracy	UR-02 & UR-04
KPI-07	Fire coverage mapping accuracy	> 90% and kept up to date	UR-02 & UR-04
KPI-08	High-risk area monitoring coverage	100% during active system operation	UR-03
KPI-09	Field maintenance of equipment completion rate	> 80% of equipment can be maintained and repaired in the field	UR-07
KPI-10	Successful implementation when using backup modes	> 90% of failures can be replaced by backup mode processes	UR-11
KPI-11	Coverage for second ignition monitoring	> 99% of surround land	UR-10 & UR-12
KPI-12	Geolocation accuracy	< 300m from detection point	Project Objective & UR-02

PRELIMINARY VERIFICATION CRITERIA

Table 17, KPI Preliminary Verification Criteria

KPI ID	METHOD	PASS CRITERIA
KPI-01	Controlled ignition test	System confirms fire before coverage exceeds 5 acres
KPI-02	Analysis of alert logs from trial	< 5% of alerts disseminated are false
KPI-03	Test completion time of system flow	Time from simulated ignition to alert transmitted < 10 min
KPI-04	Test on controlled burn with known size	System can detect a fire with a size of 25 m ²
KPI-05	Analyse autonomous classification against truthful dataset	> 90% of classifications are correct against the dataset
KPI-06	Test end to end alert transmission	> 95% of stakeholders receive a confirm fire alert with the correct location and info
KPI-07	Analysis of system mapping against reality and demonstration of continued mapping	Coverage mapped has a minimum 90% accuracy against reality and remains accurate over 60 minutes of monitoring
KPI-08	Compare coverage of sensors against high risk areas	Combined sensor coverage contains 100% of designated high risk zones
KPI-09	Inspections and analysis of equipment maintainability	Components in the field are repair per procedure > 80% of the time
KPI-10	Test by injecting a failure into the system	Backup modes can restore > 90% of functional capabilities
KPI-11	Coverage analysis	Where there is active fire monitoring, > 90% of the surround land has coverage
KPI-12	Compare system geolocation against known point	Geolocation produced by the system is within 300m of the actual location

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RISKS AND ASSUMPTIONS AFFECTING PERFORMANCE

Table 18, Risks & Assumptions Affecting System Performance

RISK	THREATENED KPIS	IMPACT	LIKELIHOOD	COUNTERMEASURES
Smoke, cloud or canopy obstructing optical sightline	KPI-03 KPI-04 KPI-07	High	High	Thermal Sensors, UAV swarm
False Detections	KPI-02 KPI-05	Medium	Low	Human confirmation, filtering out what doesn't meet the minimum heat possible for a fire
Coverage gaps from LEO satellites	KPI-03 KPI-08	High	High	Geostationary satellite feeds, UAV swarms
Loss of comms	KPI-06 KPI-03 KPI-10	High	Medium	SATCOM, onboard autonomous detection AI
Equipment fails in the field	KPI-09 KPI-10	Medium	Medium	Spare components, crew maintenance and repair training, modular designs
Autonomous detection misclassification of anomalies	KPI-05 KPI-04	High	Medium	Human confirmation, continuous model training
Geolocation error	KPI-12	Medium	Low	Sensor recalibration

We assume the existing satellite feeds remain available for the systems use, BoM allows for continued access to weather data, authorisation from airspace authorities is obtained, there is an adequate budget to fulfill this systems requirements and communication infrastructure already exists and is available.

CONCLUSION

Implementing multimodal data gathering with LEO satellites, geostationary satellites and UAVs maximises the system's coverage for locating fires as early as possible, improving the chance they can be contained before becoming a threat to stakeholders. The system outlined in this report fulfills the project objective to disseminate a fire alert containing the fires geolocation and mapping. The KPIS this system is required to meet are both measurable, and traceable to user needs, with functions being fulfilled with their allocated subsystems and unbroken data flows ensuring every part of the system is used. The approach taken to complete this report revealed system needs which aren't visible if taking a solely technical approach, outlining additional requirements such as the importance of adding thermal optics which only came to light when analysing a scenario involving poor weather conditions. The system defined in this report is a reliable preliminary basis to continue to generate next phase of the lifecycle, a detailed design, however, spread prediction and behaviour prediction require further development for improved accuracy of disseminated alerts given to the relative stakeholders.

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